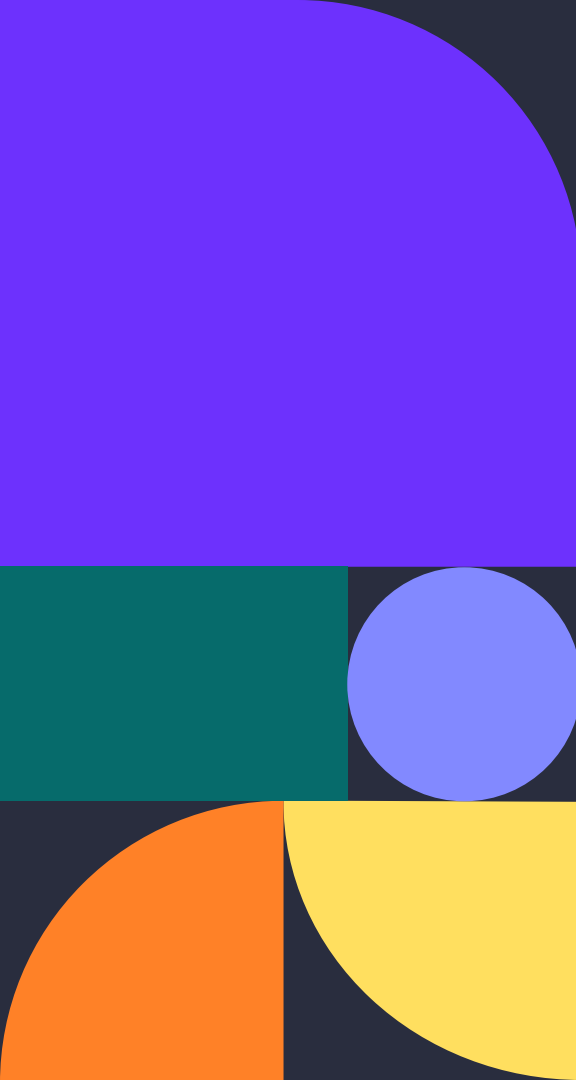




Acceptance Testing a Terraform Provider for Incus

Fabian Mettler, Bärner Go Talks 2025 no. 4, 5. November 2025

Hi, I am Fabian

- **Staff Engineering Manager** by day
- Passionate about **Open Source Software and Infrastructure Tooling**
- Contributor to the **Incus Project**
- Maintainer of **terraform-provider-incus**



Contact:

fabian@mettler.cc

<https://codeberg.org/maveonair>

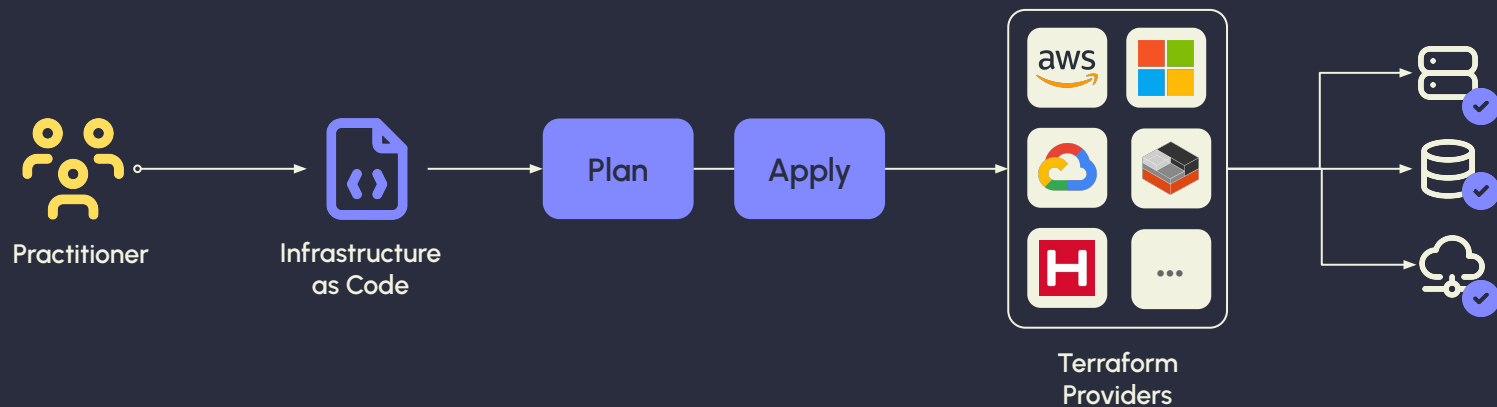
<https://github.com/maveonair>



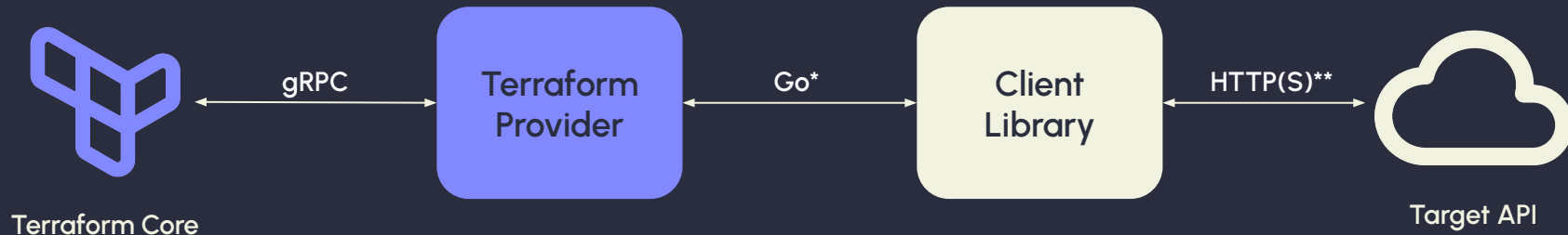
Agenda

- 01. **What is a Terraform?**
- 02. **What is Incus?**
- 03. **Acceptance Testing in terraform-provider-incus**
- 04. **Wrap up & Q&A**

What is Terraform?



What is a Terraform Provider?



* doesn't have to be written in Go

** communication doesn't need to be HTTP(S), other protocols are suitable too

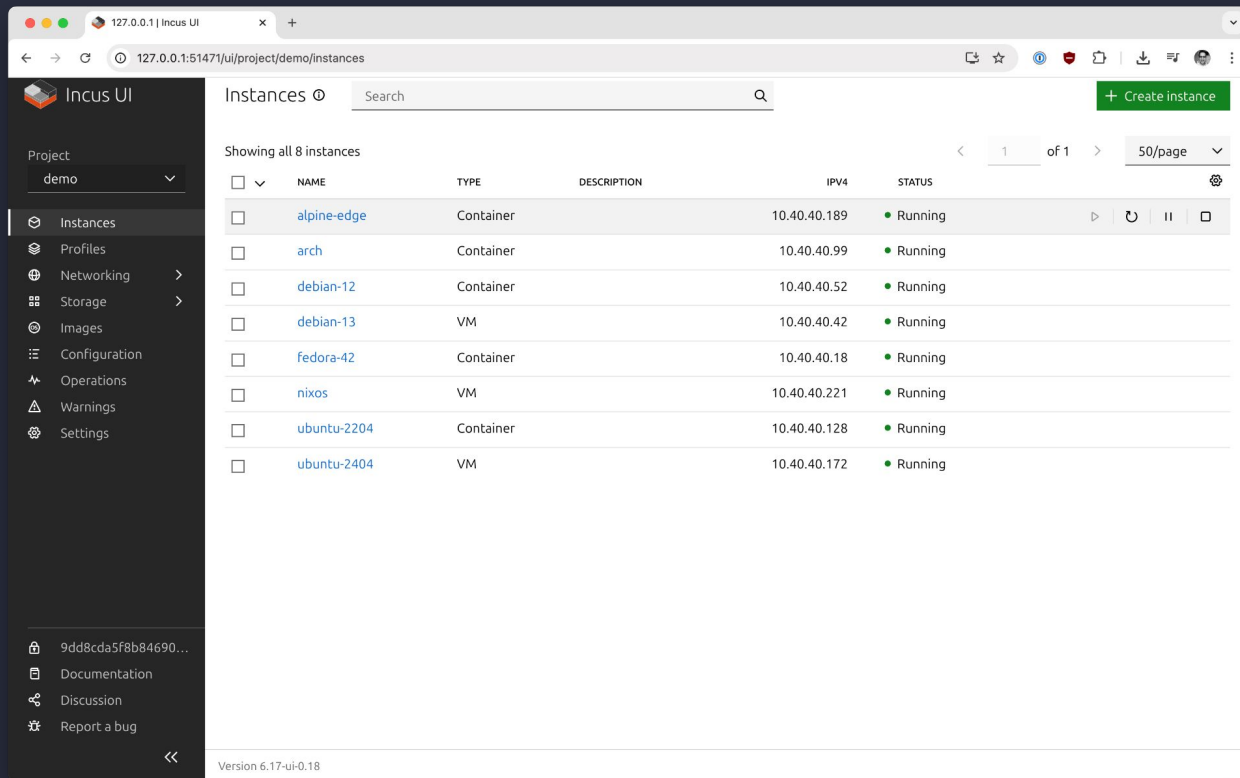
What is Incus?

```
→ ~ incus ls
```

NAME	STATE	IPV4	IPV6	TYPE	SNAPSHOTS
alpine-edge	RUNNING	10.40.40.189 (eth0)		CONTAINER	0
arch	RUNNING	10.40.40.99 (eth0)		CONTAINER	0
debian-12	RUNNING	10.40.40.52 (eth0)		CONTAINER	0
debian-13	RUNNING	10.40.40.42 (enp5s0)		VIRTUAL-MACHINE	0
fedora-42	RUNNING	10.40.40.18 (eth0)		CONTAINER	0
nixos	RUNNING	10.40.40.221 (enp5s0)		VIRTUAL-MACHINE	0
ubuntu-2204	RUNNING	10.40.40.128 (eth0)		CONTAINER	0
ubuntu-2404	RUNNING	10.40.40.172 (enp5s0)		VIRTUAL-MACHINE	0

```
→ ~ |
```

What is Incus?



The screenshot displays the Incus User Interface (UI) in a web browser. The browser's address bar shows the URL `127.0.0.1:51471/ui/project/demo/instances`. The UI features a dark sidebar on the left with the Incus logo and a navigation menu including: Project (demo), Instances, Profiles, Networking, Storage, Images, Configuration, Operations, Warnings, and Settings. At the bottom of the sidebar are links for a specific instance (9dd8cda5f8b84690...), Documentation, Discussion, and Report a bug. The main content area is titled 'Instances' and includes a search bar and a '+ Create instance' button. It shows 'Showing all 8 instances' with a table listing them. The table has columns for NAME, TYPE, DESCRIPTION, IPV4, and STATUS. All instances are currently in a 'Running' state. The footer of the UI indicates 'Version 6.17-ui-0.18'.

NAME	TYPE	DESCRIPTION	IPV4	STATUS
alpine-edge	Container		10.40.40.189	Running
arch	Container		10.40.40.99	Running
debian-12	Container		10.40.40.52	Running
debian-13	VM		10.40.40.42	Running
fedora-42	Container		10.40.40.18	Running
nixos	VM		10.40.40.221	Running
ubuntu-2204	Container		10.40.40.128	Running
ubuntu-2404	VM		10.40.40.172	Running

terraform-provider-incus

```
resource "incus_instance" "alpine_edge" {
  name = "alpine-edge"
  image = "images:alpine/edge"

  config = {
    "boot.autostart" = true
    "limits.cpu"     = 2
  }
}
```



```
~/C/g/m/t/terraform-provider-incus-acctest
+ terraform-provider-incus-acctest tofu plan

OpenTofu used the selected providers to generate the following execution plan. Resource actions are indicated
with the following symbols:
+ create

OpenTofu will perform the following actions:

# incus_instance.alpine_edge will be created
+ resource "incus_instance" "alpine_edge" {
  + architecture = (known after apply)
  + config       = {
    + "boot.autostart" = "true"
    + "limits.cpu"    = "2"
  }
  + ephemeral    = false
  + image        = "images:alpine/edge"
  + ipv4_address = (known after apply)
  + ipv6_address = (known after apply)
  + mac_address  = (known after apply)
  + name         = "alpine-edge"
  + profiles     = [
    + "default",
  ]
  + running      = true
  + status       = (known after apply)
  + target       = (known after apply)
  + type         = "container"
}

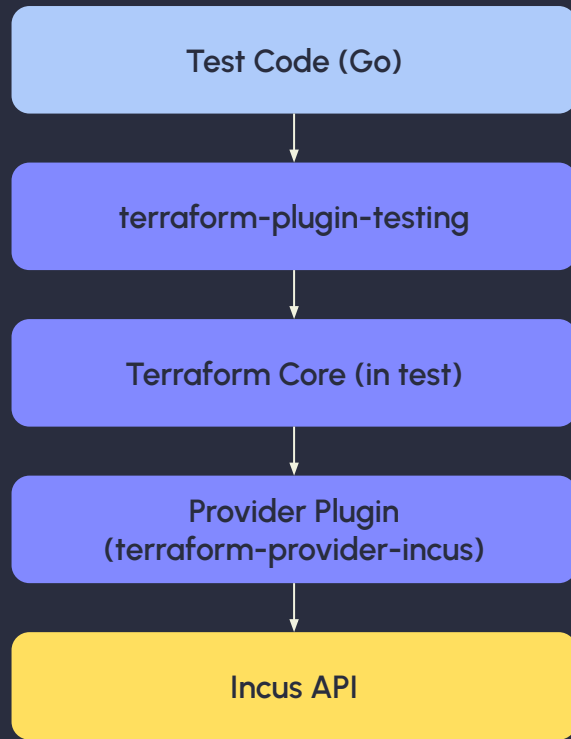
Plan: 1 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so OpenTofu can't guarantee to take exactly these
actions if you run "tofu apply" now.
+ terraform-provider-incus-acctest
```

Demo

Terraform Acceptance Testing in Incus

- Acceptance tests run real Terraform operations against the real Incus APIs
- Used to verify provider functionality end-to-end
- Implemented using the `terraform-plugin-testing` framework
- Tests create, update, and destroy real resources
- Run with `TF_ACC=1`



Terraform Acceptance Testing in Incus



```
func TestAccInstance_Basic(t *testing.T) {
    resource.Test(t, resource.TestCase{
        ProtoV6ProviderFactories: testProviderFactories,
        Steps: []resource.TestStep{
            {
                Config: `resource "incus_instance" "test" { name = "tf-acc" ... }`,
                Check: resource.TestCheckResourceAttr("incus_instance.test", "name", "tf-acc"),
            },
        },
    })
}
```

Challenges of Acceptance Testing with Incus

GitHub Actions limitations

- Runs inside OCI containers → limited kernel features
- No VMs (no /dev/kvm), no clusters
- Restricted networking (bridges, VLANs, etc.)
- Limited storage features (only local storage backends)
- Occasional flaky tests due to containerized setup

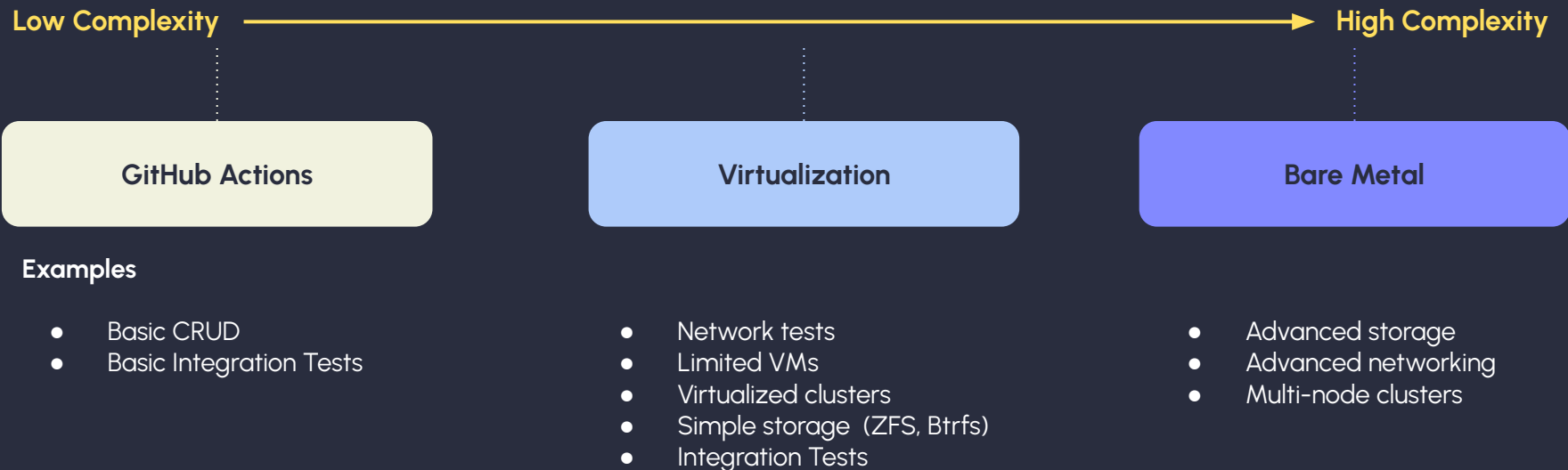
Infrastructure needs

- Local test environment required for VMs, clusters, advanced networking, and storage features

Operational challenges

- Must reset Incus between tests
- Concurrency issues with shared resources

My test infrastructure



Demo

Q&A